ECVOL-bench: An Open, Identity-Controlled Benchmark for Multimodal Earnings-Call Volatility Prediction

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Abstract

A decade of work predicts post-earnings-call stock volatility from call transcripts and audio. We show that the evaluation substrate this literature shares cannot support its claims: the canonical label and split files have a zero-day embargo against 30-day targets, place 44–80% of test calls' companies in training, and anchor day 0 on the call date although the earnings release precedes the open for most calls. We release ECVOL-bench, an open benchmark over FinCall-Surprise and MAEC that repairs this: reconstructed company identity and dates for a corpus that ships without them, EDGAR-measured release timestamps, targets under both horizon conventions, embargoed and ticker-disjoint splits, econometric baselines, and an identity-control suite that runs on any prediction file. Under these controls, modern frozen text and audio representations add nothing to past volatility: adding either to a past-volatility ridge lowers R^2_{OOS} in all 16 split×horizon cells, text embeddings identify the company at 89.5% probe accuracy, audio-only heads do no better than a featureless intercept, and fusion beats HAR-RV in one horizon. Re-run on their own data, the field's text models lose 12–82% MSE when companies are held out while price-based models do not move. Nothing degrades when the frozen pipeline is scored on 475 calls held after every encoder's knowledge cutoff. All code, metadata, splits, targets, features and result tables are released.

1 Introduction

Earnings calls are the one recurring event at which management speaks at length, semi-spontaneously, about a public company, in both *what* is said and *how*. From Qin and Yang (2019) through HTML (Yang et al., 2020), MAEC (Li et al., 2020), VolTAGE (Sawhney et al., 2020a), DialogueGAT (Sang and Bao, 2022), KeFVP (Niu et al., 2023) and ECC Analyzer (Cao et al., 2024), a lineage of multimodal models reports beating classical methods at predicting the 3–30-day log realised volatility that follows a call. Two observations undercut that progress: Yu et al. (2025) show that transcript representations mainly encode *which company* is speaking and that training-free baselines on a company's own volatility history beat every transcript model, and only ∼14% of reported financial-NLP results reproduce (Tatarinov et al., 2025).

We argue both are properties of the *evaluation substrate* the field shares, and that it can be repaired. Nearly every paper above scores on one physical set of label and split files passed from paper to paper. Those files have no embargo between training and test although targets run 30 days past each call; most test calls come from companies in training, and no paper reports the disjoint condition; they carry degenerate cells; they encode a calendar-day horizon that downstream work describes as trading days; and they anchor day 0 on the call date although, as we measure from EDGAR, the earnings release precedes the market open for most calls. The open corpus with raw audio, FinCall-Surprise (Shu et al., 2026a), ships with no ticker, date or speaker metadata, so the identity controls that would expose the confound cannot be run on it; its successor MM-FinEval (Shu et al., 2026b) adds ticker and date fields but no embargoed or company-disjoint split, past-volatility baseline, or identity control.

Contributions. (1) **ECVOL-bench**: identity and dates reconstructed for FinCall-Surprise (92.9% of calls, 60/60 manual audit), release timestamps measured from 8-K filings, three target families under both horizon conventions, embargoed temporal and ticker-disjoint splits, SHA-256 manifests, and a harness in which every table regenerates from one command (§2). (2) **A substrate audit** of the shared label and split files (Table 1). (3) **An identity-control suite** (ticker probes, same-company and global content shuffles, ticker-disjoint evaluation), released as a command that audits *any* prediction file. (4) **What the controls show** for modern open-weight text and audio representations against strong econometric baselines (§4). (5) **A reproduction study** of the lineage's five runnable models on their own data under the same controls, with a code-availability audit of seventeen papers. (6) **Post-cutoff evaluation** of the frozen pipeline on 475 calls from late 2025. Stages, controls and confirmatory tests were pre-registered in a public decision log, including the positive-framing gate the content models missed.

Related work. Known pitfalls organise the design: ticker-identity leakage (Yu et al., 2025), which LLMs reproduce even on entity-masked transcripts (Lopez-Lira et al., 2025); LLM lookahead bias, with diagnostics and cutoff-bounded models (Gao et al., 2025; Yan et al., 2026; Benhenda, 2026) but no frozen earnings-call

Benchmark / split	calls	overlap	embargo	label defect
EC (Qin & Yang)	560	80.4%	0 / 0 d	137 zeros
MAEC-15 (KeFVP files)	765	44.2%	0 / 0 d	binary cols
MAEC-16 (KeFVP files)	1400	55.7%	0 / 3 d	binary cols
ours: FinCall temporal	2200	91.3%	47 / 47 d	—
ours: FinCall ticker-disjoint	2433	**0.0%**	n/a	—
ours: MAEC temporal	2264	79.4%	101 / 48 d	—
ours: MAEC ticker-disjoint	2688	**0.0%**	n/a	—

Table 1: Substrate audit of the shared benchmark files (Result Table 5R, regenerated by `ecvol audit substrate` from SHA-256-manifested copies of the VolTAGE and KeFVP repositories, which are byte-identical). Overlap is the share of test calls whose company is in training; embargo is train→val / val→test. Every temporal split permits overlap, ours included; no paper in the lineage reports the disjoint condition. Shipped MAEC labels join our calendar-day, call-date-anchored targets on 77–79% of test calls at $\tau \geq 7$ and correlate 0.52–0.87 with them.

pipeline evaluated post-cutoff; weak baselines, where persistence, GARCH and HAR-RV (Corsi, 2009) are hard to beat and persistence is the natural denominator (Gu et al., 2020); and gender bias in audio features (Wang et al., 2024; Lin et al., 2024). Yu et al. (2026) time releases from 8-Ks for return prediction; no prior work audits the volatility label sets. The strongest recent positive claims (Chen et al., 2025; Cao et al., 2024) report no identity controls and release no code.

2 The Substrate and the Benchmark

The substrate the literature shares. The EarningsCall (EC) benchmark of Qin and Yang (2019) and the two MAEC cohorts are scored on split and label files redistributed by VolTAGE and KeFVP. We fetched them, manifested them by SHA-256 and audited them (Table 1; `ecvol audit substrate`). The VolTAGE and KeFVP test files are byte-identical: one physical label set, never recomputed from prices. Every train/validation and validation/test boundary has a zero-day gap (on EC, training ends and validation starts on 2017-08-03) against targets that run 30 days past each call. On the temporal splits the field reports, 80.4% (EC), 44.2% (MAEC-15) and 55.7% (MAEC-16) of test calls come from companies also in training; a temporal split always permits this, but no paper in the lineage reports the disjoint condition, so the confound of Yu et al. (2025) has never been measured on these benchmarks. The source paper defines its horizons over *calendar* days; most downstream work describes them as trading days, which at τ=30 is 21 versus 30 sessions. Finally, the shipped MAEC labels correlate only 0.52–0.87 with calendar-day targets recomputed from prices under the same call-date anchor (0.25–0.66 at τ=3), so the lineage's labels are not reproducible from prices under any single documented rule.

Corpora and identity. **FinCall-Surprise** (Shu et al., 2026a) (primary; Apache-2.0; 2,688 calls, 2019–2021, MP3 audio, transcripts, slides) ships four fields per record and no ticker, date, CIK or speaker names. We reconstruct identity and date from transcript content and slide metadata, cross-check against prices, and release the table with an 80-row manual override list: 2,496 calls resolve (92.9%; 95.3% of earnings-type calls) and a 60-call manual audit passes 60/60. **MAEC** (Li et al., 2020) (secondary; CC-BY-SA-4.0; ∼3,443 calls, 2015–2018, sentence-aligned audio features, no raw audio) and **Earnings25** (Jiang et al., 2026) (CC-BY-4.0; 514 S&P 500 calls held July–December 2025, webcast MP3 at 64 kbps for 281 calls and ≤24 kbps for 232) complete the set; Earnings25 is post-cutoff for every model we use, and 475 of its calls resolve to a constituent ticker and join prices.

Targets. Daily adjusted prices come from a free source cross-checked against Tiingo on 50 tickers. With r_t the return on session t relative to day 0, the last session before the call's information is public, the primary target is $v_{\text{post}}(\tau) = \ln \sqrt{\frac{1}{n}\sum_{t=1}^{n}(r_t - \bar{r})^2}$ for $\tau \in \{3, 7, 15, 30\}$, with $v_{\text{pre}}(\tau)$ defined over the sessions ending at day 0. Under the **trading-day** convention $n = \tau$; under the **calendar-day** convention of Qin and Yang (2019) the window holds every session within τ calendar days, so $n \approx 0.7\tau$ (FinCall: 2.4/4.9/10.8/21.2 sessions) and at τ=3 at most one session for a Thursday or Friday call. Windows with fewer than two sessions are excluded with a reason code (42.5% of FinCall's calls at τ=3 calendar days, for which the released label sets still carry a value). The conventions are far from interchangeable: v_{post} correlates 0.78 between them at τ=3 and the persistence MSE, the leaderboard's metric, is 44–63% higher there under calendar days, so a published MSE is uninformative unless its convention is stated (Appendix Table 9). Both ship as separate files; every result below uses trading days, and the convention is a column of every result table. Two identity-robust variants are added: the **change** $\Delta v = v_{\text{post}} - v_{\text{pre}}$, which subtracts the company's own level, and the **HAR-residual**. After joining, FinCall yields 2,433 calls with a target (97.5%) and MAEC 2,688 (78.6%; 2015–2018 delistings).

Measured call timing. Fewer than 4% of FinCall transcripts mention a clock time, so no call carries a measured time of day. We take the information boundary from EDGAR: the acceptance time of the earliest 8-K with Item 2.02 within three days before to one day after the call date (89.3% of FinCall, 68.6% of MAEC, 99.8% of Earnings25), with recorded fallback tiers (Earnings25 call-start time, the before/after-market flag of Yu et al. (2025), an after-hours assumption). A release before the open makes the call date itself day 1. Against the call-date anchor that every released label set uses, **day 0**

moves for 53% of FinCall, 56% of MAEC and 72% of Earnings25 calls, almost always one session earlier; under the published anchor the reaction session sits in the *pre* window for most of the corpus. On the measured anchor the persistence MSE is up to 11% higher on FinCall, 25% on MAEC and 27% on Earnings25, and because the reaction session now enters the post window, Δv acquires a positive mean (FinCall $+0.52$ at τ=3, $+0.10$ at τ=30). The call-date targets are shipped as a labelled variant for comparison with prior label sets.

Splits and release. *Temporal* splits (∼70/10/20, calendar order) carry a 30-trading-day embargo so that no target window crosses a boundary; *ticker-disjoint* splits (grouped, seeded) put no validation or test company in training (overlap 0.0%, against 91.3% FinCall and 79.4% MAEC on our own temporal splits); a *combined* split enforces both. Every feature function receives an `as_of` timestamp and the harness fails any feature that reads later price data; leakage assertions run in CI. The release contains every artefact above, the derived feature archives, the harness, and every result CSV with its run manifest (resolved config, git SHA, seeds, file hash); raw audio and transcripts are never redistributed.

3 Models, Protocol and Controls

Ladder. **Stage 0**: persistence ($\hat{v} = v_{\text{pre}}$), a train-mean intercept, EWMA (λ=0.94), HAR-RV (Corsi, 2009) and GARCH(1,1). **Stage 1**: LightGBM over past-volatility features, transcript length, Q&A turn count and a **ticker fixed effect**, the in-house identity baseline. **Stage 2**: transcripts segmented into prepared remarks, Q&A and speaker turns; BGE-M3 (BAAI, 2024) section-pooled embeddings (1024-d), FinBERT (ProsusAI, 2019) sentiment aggregates and surface statistics. **Stage 3**: audio resampled to 16 kHz mono; eGeMAPSv02 functionals (Eyben et al., 2010), WavLM-Large (Chen et al., 2021) and emotion2vec+ (Ma et al., 2023) embeddings over 30 s windows, mean-pooled per call. **Stage 4**: gated fusion and a late-fusion stack (meta-ridge over the text, audio and Stage-1 predictions, fit on validation). From Stage 2 upward every head is reported with and without a past-volatility block (v_{pre} at all horizons plus HAR-RV features); heads are ridge (α selected on validation) and a shallow MLP (5 seeds). The MLP heads are seed-unstable (across-seed R^2_{OOS} std up to 6.8), so the grid uses one pre-registered deterministic canonical per stage; MLP cells are in Appendix A.

Metrics and tests. We report MSE on log-RV and, as the headline, R^2_{OOS} vs. persistence (Gu et al., 2020), $1 - \sum(y - \hat{y})^2 / \sum(y - \hat{y}_{\text{persist}})^2$. Because Δv has a positive mean on the measured anchor, persistence (Δv=0) is a weak floor there, so every Δv claim is also stated against the train-mean intercept. Diebold–Mariano tests (Diebold and Mariano, 1995) with the HLN correction compare per-call squared errors against persistence and the preceding stage; confirmatory families (Stage-k vs. Stage-1 on Δv across horizons; Stage 4 vs. best unimodal; published vs. controlled split per reproduced model) are Holm-corrected. Per-year, COVID-exclusion, section and gender analyses are exploratory.

Identity-control suite. (1) A *ticker-only* model predicts each company's training mean. (2) A *same-company shuffle* gives each test call the transcript or audio of another call of the same company; a *global shuffle* gives it any other call. A model unchanged under the former reads identity; one unchanged under the latter reads nothing. (3) A linear *identity probe* predicts the ticker from frozen call embeddings. (4) ticker-disjoint results accompany temporal ones everywhere. `ecvol audit predictions` applies the prediction-space forms of these controls, plus split integrity, seen-vs-unseen companies, the between-company share of prediction variance and company-clustered significance, to any file of (call, ticker, date, split, horizon, truth, prediction) rows, so a published result can be audited from the one artefact its authors already have.

Reproductions and post-cutoff tests. Of seventeen papers in the lineage, one releases code that runs as released (Same-Company-Same-Signal), seven ship code needing repairs or unshipped inputs, and nine ship nothing usable (Table 15). We re-run the five with usable code (HTML, KeFVP, Same-Company-Same-Signal, Sawhney et al., 2020b, DialogueGAT) *on their own data*, with the authors' code where it can be made to start and a line-by-line port where it cannot (every patch in a released ledger), first against the published number, then under an embargoed and a ticker-disjoint split of the same data. Earnings25 supports two further tests: the frozen Stage-3 ladder by bitrate stratum, and the entire FinCall-trained pipeline scored, without retraining, on calls held after every encoder's knowledge cutoff.

4 Results

Headline tables use FinCall-Surprise, the identity-robust Δv target and the trading-day convention; MAEC, level-v, MSE, both conventions and every head are in Appendix A, regenerated byte-identically by `ecvol report` from manifested CSVs.

The floor. Table 2 sets it. HAR-RV is strong except in one cell: on the FinCall temporal split at τ=30 the training window is COVID-inflated and the test year (2021) is calm, so every long-horizon level forecast, ours and every content head, falls below persistence (level-v: HAR -0.29, text -0.45, audio -0.76, fusion -0.23). The same models pass on the ticker-disjoint split and on MAEC (a documented regime exception).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
S0 train mean	0.165	0.158	0.060	0.088
S0 HAR-RV	0.444	0.372	0.138	−0.285
S0′ past-vol ridge	0.412	0.314	0.094	0.113
S1 GBDT (ticker FE)	0.340	0.249	−0.126	−0.948
S2 text ridge + vol	0.333	0.285	0.080*	−0.077*
S3 WavLM ridge + vol	0.353	0.268	0.033*	0.050*
S4 stack fusion	0.406*	0.347*	0.066*	0.100*
ticker-disjoint split (n=484)				
S0 train mean	0.200	0.201	0.132	0.035
S0 HAR-RV	0.570	0.491	0.349	0.345
S0′ past-vol ridge	0.552	0.469	0.345	0.318
S1 GBDT (ticker FE)	0.494	0.440	0.292	0.237
S2 text ridge + vol	0.399†	0.375	0.256	0.301
S3 WavLM ridge + vol	0.442	0.364	0.241	0.217
S4 stack fusion	0.531	0.465	0.340	0.396*

Table 2: The ladder on FinCall-Surprise. R^2_{OOS} vs. persistence on Δv (test); persistence is 0 by construction, the train-mean intercept is the featureless floor. One pre-registered canonical model per stage; $*$/†: Holm-corrected Diebold–Mariano better/worse than Stage 1 ($p < 0.05$). Adding text or audio to the past-volatility ridge (S0′) lowers R^2_{OOS} in every cell. Full grid, MAEC, level-v, MSE and MLP heads: Appendix A.

Identity probe (linear, frozen call embedding)	accuracy	×chance
BGE-M3 text, FinCall (357 tickers)	89.5%	319×
BGE-M3 text, MAEC (902 tickers)	53.9%	487×
WavLM audio, FinCall (356 tickers)	76.3%	271×
emotion2vec+ audio, FinCall (356 tickers)	29.4%	105×

Content shuffle (R^2_{OOS} on Δv)	τ=3	τ=7	τ=15	τ=30
Text ridge, temporal split				
real	0.182	0.202	0.044	−0.133
same-company shuffle	0.174	0.194	0.047	−0.110
global shuffle	0.145	0.110	0.008	−0.152
Text ridge, ticker-disjoint split				
real	0.195	0.217	0.195	0.169
same-company shuffle	0.176	0.175	0.076	−0.322
global shuffle	0.161	0.128	0.016	−0.410
WavLM ridge + past-vol, ticker-disjoint split				
real	0.456	0.390	0.267	0.222
same-company shuffle	0.453	0.373	0.229	0.152
global shuffle	0.453	0.323	0.164	0.099

Table 3: Identity controls. Top: frozen embeddings identify the company far above chance. Bottom: giving each test call another call of the *same* company changes the text head little where companies recur (temporal), while a global shuffle degrades it; on the ticker-disjoint split the text head's residual call-specific signal at $\tau \geq 15$ sits below the past-volatility floor of Table 2, and the audio head's apparent gain survives a global shuffle at τ=3.

Because the reaction session enters the post window on the measured anchor, a featureless train-mean intercept already reaches 0.17/0.16/0.06/0.09 (temporal) and 0.20/0.20/0.13/0.04 (ticker-disjoint), the reference for every Δv claim below.

Text reads identity (Table 3). The text-only ridge head sits at the intercept: 0.18/0.20/0.04/−0.13 (temporal) and 0.20/0.22/0.20/0.17 (ticker-disjoint), never within 0.09 of HAR-RV; the MLP heads overfit. Adding text to the past-volatility ridge *lowers* R^2_{OOS} in all eight cells. A linear probe recovers the ticker from the frozen call embedding at 89.5% on FinCall and 53.9% on MAEC. On the temporal split, where test companies recur in training, giving each test call the transcript of *another call of the same company* changes the head by at most 0.02 while a global shuffle degrades it: the head reads which company is speaking. On the ticker-disjoint split the same-company swap does cost the head at $\tau \geq 15$, so some call-specific signal exists in unseen companies, but it lies below the past-volatility floor and vanishes once that block is present (text+vol 0.301 vs. past-vol 0.318 at τ=30). MAEC repeats the pattern.

Audio is inert beyond past volatility. Audio-only ridge heads on eGeMAPS, WavLM-Large and emotion2vec+ sit within 0.10 of the train-mean intercept at every horizon on both splits and never above it by more than 0.02 (temporal WavLM 0.156/0.148/−0.027/−0.008 against 0.165/0.158/0.060/0.088). Adding WavLM to the past-volatility ridge lowers R^2_{OOS} in all eight cells. The audio-shuffle control (Table 3, bottom) shows the past-vol-augmented head keeping its score when every test call receives a *random other company's* audio at τ=3 (0.456 → 0.453); at longer horizons the shuffle costs more, but the real-audio head is still below the past-vol-only ridge, so what audio contributes does not add to past volatility. Identity is nonetheless encoded: a probe recovers the ticker from pooled WavLM embeddings at 76.3%. An exploratory gender check (dominant-speaker F0 proxy) finds per-group test MSE of 0.243 vs. 0.239. Re-run on Earnings25 by bitrate (one call per company, so the global shuffle is the control), the audio increment (real minus shuffled R^2_{OOS}) never exceeds 0.03, is negative in 7 of 12 stratum cells, and the 64 kbps stratum is not the better one (Appendix Table 13): the FinCall result is not an artefact of its 40 kbps transcodes.

Fusion. Gated fusion is worse than its inputs everywhere. The late-fusion stack is the best content stage but stays below HAR-RV in six of eight cells; it exceeds HAR-RV only in the regime cell (0.100 vs. −0.285, still below the past-vol ridge at 0.113) and at ticker-disjoint τ=30 (0.396 vs. 0.345; Holm p=0.003 vs. Stage 1, no pre-registered test vs. HAR-RV). The stack beats the text and audio stages in seven of eight cells but not the past-volatility stage, so multimodal is not better than the best unimodal model once past volatility is that model, and the pre-registered positive-framing gate (content gains surviving the controls on at least two of four horizons) was missed at Stages 2, 3 and 4, as the decision log records.

The field's own models (Table 4). Of the five runnable papers, the results that reproduce are those of price-only or same-company models (Appendix A.3). Same-Company-Same-Signal reproduces from the shipped data: TSMixer to 10^{-5}, TMLP within 0.02 including the paper's finding that embeddings shuffled within a company do as well as the real ones, and its training-free augmented same-company history lands within 0.05 of KeFVP's published MSE at every horizon on EC and MAEC-15 with no text and no training, a model with no definition once companies are held out. KeFVP, a price series conditioned on text whose audio branch is disabled in its own code, reproduces seven of eight MAEC cells within one published standard deviation after six import-level repairs; its EC row, run without the unavailable knowledge-enhanced embeddings, sits above all four published cells and is not counted. HTML's released driver splits features and labels with independent unseeded shuffles, so it cannot have produced the published table; with the authors' model classes verbatim the published recipe underfits at every horizon ≥ 15 on every split. The published ensemble of Sawhney et al. (2020b) is reachable only because untrained text and audio branches act as an intercept on a finance SVR whose mixing weights the code tunes on the test set. DialogueGAT's corpus was never released. Under our controls (Table 4), holding companies out costs the three text models 12–82% MSE while TSMixer and KeFVP do not move, with training sets as large (EC) or reduced by the same third (DEC, MAEC); the embargo moves models by at most 13%; and HTML's own trained model changes by at most 4% when a test transcript is swapped for another call of the same company (under 1% on the ticker-disjoint split) against 9–23% for a different company's call. What HTML reads from a transcript is which company is speaking.

Model (data), change in test MSE	τ=3	τ=7	τ=15	τ=30
Companies held out (ticker-disjoint vs. the model's own split)				
HTML text (EC)	+18%	+55%	+55%	+82%
HTML text+audio (EC)	+20%	+44%	+61%	+70%
Sawhney ensemble, val-tuned (EC)	+13%	+28%	+41%	+38%
SCSS TMLP, text (DEC)	+12%	+23%	+25%	+27%
SCSS TSMixer, price (DEC)	+7%	+5%	−6%	−8%
KeFVP, text+price (MAEC-15)	−3%	+4%	+3%	−1%
KeFVP, text+price (MAEC-16)	−9%	−3%	−10%	−4%
30-session embargo vs. the model's own split				
HTML text (EC)	+10%	+3%	−3%	+11%
Sawhney ensemble (EC)	+5%	−6%	−2%	−9%
SCSS TMLP (DEC)	−4%	+1%	0%	+3%
HTML text, transcript replaced by (own split, 97 calls)				
another call, *same* company	+1%	+4%	+3%	+2%
a call of a *different* company	+9%	+11%	+21%	+23%

Table 4: The field's runnable models under our controls (Result Table 6R-C). Each row keeps the configuration of Appendix Table 11 and changes only the split. Holding companies out costs every text model 12–82% and leaves the price-dominated models unchanged, with held-out training sets as large (EC) or cut by the same third (DEC, MAEC). Absolute values: Appendix Table 12.

Head (in-cutoff → post-cutoff)	τ=3	τ=7	τ=15	τ=30
S0 train mean	0.36→0.58	0.14→0.47	−0.37→0.15	−1.05→−0.29
S0 HAR-RV	0.44→0.60	0.37→0.54	0.14→0.35	−0.29→0.08
S2 text ridge + vol	0.43→0.57	0.34→0.50	0.03→0.25	−0.45→−0.20
S3 WavLM ridge + vol	0.38→0.57	0.24→0.43	−0.09→0.15	−0.66→−0.32
S4 fusion ridge + vol	0.41→0.50	0.32→0.46	−0.02→0.20	−0.63→−1.93

Table 5: Frozen-pipeline post-cutoff evaluation (Result Table 7). Heads fit once on FinCall's training split, scored unchanged in-cutoff (the 2021 FinCall test year) and post-cutoff (475 Earnings25 calls, late 2025); level-v, R^2_{OOS} vs. persistence. No head competitive in-cutoff degrades; the ordering survives, with HAR-RV above every content head at every horizon. Month-cluster bootstrap CIs and Δv rows are in the released CSV.

Post-cutoff (Table 5). Scored unchanged on 475 Earnings25 calls from late 2025, after the training cutoff of every encoder and on companies none of the heads has seen, no head that was competitive in-cutoff degrades on level-v: HAR-RV rises from 0.44/0.37/0.14/−0.29 to 0.60/0.54/0.35/0.08 and the content ridges rise with it (the 3k-dimensional fusion ridge is unstable at τ=30). Levels are not comparable across the cutoff (different years and universes); the ordering is, and it survives: HAR-RV beats every content head at every horizon, and no content head that was competitive in-cutoff collapses post-cutoff. Under the call-date anchor of the published label sets the same frozen heads collapse in-cutoff (text+vol on Δv: -1.7 to -3.6; Appendix Table 14), because the learned past-volatility block absorbs the reaction session that the after-hours rule leaves in the pre-window.

5 Discussion and Conclusion

Four statements hold on the completed ladder. Strong econometric baselines are hard to beat, and R^2_{OOS} vs. persistence exposes a regime failure that raw MSE hides. Frozen text content adds nothing over past volatility, for a measurable reason: call embeddings identify the company, and where companies recur a same-company transcript swap leaves the head unchanged. Frozen audio content is inert beyond past volatility on 40 kbps transcodes and 64 kbps webcasts alike, while still encoding the speaker's company. Fusion does not rescue the modalities it fuses. Each statement depends on apparatus the shared substrate lacks. Applied to the field's own code and data, that apparatus returns the same diagnosis: what reproduces is price-only or same-company history, and the text models lose their advantage when the company is unseen. The strongest recent positive claims (Chen et al., 2025; Cao et al., 2024) report no identity controls and release no code; `audit predictions` makes the check cheap for any author with a prediction

file. The verdict is not that transcripts and audio carry no information about a company's risk, but that the benchmarks this literature uses cannot tell that information apart from knowing which company is speaking, and that once they can, the past-volatility floor has not been beaten.

Limitations

One post-cutoff quarter. The lookahead test rests on 475 calls from a single calm quarter and a different universe from the training corpus, so it bounds contamination in the ordering of models, not in their absolute scores. **Reproductions run on their own data**, not on ECVOL-bench, to keep "the model fails" separable from "the model does not transfer"; two carry known costs (KeFVP's released EC embeddings are unavailable upstream; DialogueGAT's corpus was never released and our port stops at its protocol). **MLP heads are seed-unstable** (across-seed R^2_{OOS} std up to 6.8 on level-v, 1.1 on Δv), which is why the grid uses deterministic canonicals; MLP cells in the appendix are seed means with that caveat. **Short-horizon R^2_{OOS} inflation.** The persistence denominator is noisiest at τ=3 (MSE $\sim$2.8$\times$ that at τ=7), so short-horizon gains are easier to show. **Timestamps.** 10.7% of FinCall and 31.4% of MAEC calls have no Item 2.02 filing in the window and keep a flagged after-hours fallback; intraday releases are treated as after-hours; results under the call-date anchor of prior label sets are not interchangeable with ours and are shipped as a variant. **Regime sensitivity.** FinCall-Surprise spans COVID; the temporal τ=30 failure shows long-horizon level forecasts are regime-sensitive, which is why we emphasise Δv and per-year breakdowns. **MAEC coverage.** Its 78.6% price-join rate (2015–2018 delistings) makes it a secondary corpus, and it ships no raw audio. **Source audio.** Every FinCall file is a 40 kbps MP3 transcode and Earnings25 is 64 kbps or lower, so no result here is established on lossless audio. **No diarisation**; audio is pooled per call, and the gender analysis uses a dominant-speaker F0 proxy. **Identity reconstruction is imperfect**: 7.1% of FinCall calls remain unresolved and are excluded with a reason code.

Ethics Statement

Data and licensing. We use openly licensed corpora (FinCall-Surprise, Apache-2.0; MAEC, CC-BY-SA-4.0; Earnings25, CC-BY-4.0) and open-weight models; derived features inherit and state their upstream license, and raw audio or transcripts from sources with unclear terms (EC, DEC, the reproduced repositories) are never redistributed. Calls are public corporate disclosures; we release no speaker-level audio or identity, only pooled per-call features. **Gender confound.** Audio features encode speaker gender and female executives are underrepresented in these corpora (Wang et al., 2024; Lin et al., 2024; Mathur et al., 2022); we report a gender-confound diagnostic rather than silently propagate the bias. **Intended use.** This is a forecasting-research benchmark, not investment advice; predictions are noisy and regime-sensitive and should not drive automated trading. **Reproducibility.** Given the documented reproducibility crisis in financial NLP (Tatarinov et al., 2025), the environment is pinned, every data file is checksummed, and every result table regenerates byte-identically from committed artifacts with a run manifest.

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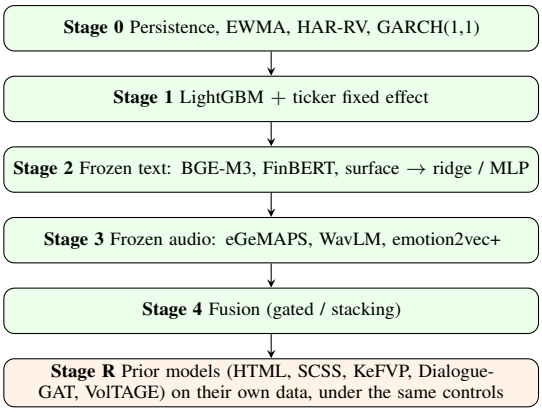


Figure 1: The modelling ladder. From Stage 2 up, every model also carries past-volatility covariates (and is reported with them ablated). Every stage runs through the identity-control suite. Green = complete, orange = pending.

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A Extended Results

This appendix holds the full head set behind each headline table, the horizon convention delta, the reproduction study in detail, the Earnings25 bitrate study, the anchor-sensitivity variant of the post-cutoff test, the code-availability audit, and the complete generated grids. Every table regenerates from a committed result CSV with a run manifest (`ecvol report`; `REPRODUCE.md` lists one command per table).

A.1 Stages 0–3: all heads

Tables 6, 7 and 8 give the full Stage-0/1, Stage-2 and Stage-3 head sets on the Δv target, including the past-vol-only and MLP heads omitted from Table 2.

A.2 Horizon convention

A.3 Reproduction study

Table 10 is the first entry of the study: the HTML head over our frozen BGE-M3 chunk sequences on EC under the published split, an embargoed split and a ticker-disjoint split with identical labels. Table 11 re-runs the five papers with usable code on their own data with the authors' code wherever it runs and a line-by-line port where it cannot; the released fidelity ledger records every patch and substitution. Table 12 gives the absolute values behind Table 4, and Table 15 the code-availability audit of the seventeen papers.

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.340*	0.249*	-0.126	-0.948*
Ticker-disjoint split (n=484)				
Persistence	0.000	0.000	0.000	0.000
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.494*	0.440*	0.292*	0.237*

Table 6: Baselines. R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). HAR-RV is strong except on the COVID-regime temporal τ=30 cell. $*$: Diebold–Mariano significant vs. persistence ($p < 0.05$).

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (text)	0.182*	0.202	0.044*	-0.133*
Ridge (past-vol)	0.412*	0.314*	0.094*	0.113*
Ridge (text+vol)	0.333	0.285	0.080*	-0.077*
MLP (text)	0.013*	0.069*	-0.253	-1.202
Ticker-disjoint split (n=484)				
Ridge (text)	0.195*	0.217*	0.195*	0.169
Ridge (past-vol)	0.552*	0.469	0.345	0.318*
Ridge (text+vol)	0.399*	0.375	0.256	0.301
MLP (past-vol)	0.577*	0.508*	0.381*	0.356*

Table 7: Text content (Stage 2). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). The text-only head does not separate from the past-volatility head; adding text to past-vol does not help. $*$: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$). Full head set in Appendix A.

Same-Company-Same-Signal reproduces: its TSMixer baseline matches the 76 published cells to a median 10^{-5} from the shipped data alone, its TMLP text model on the authors' OpenAI embeddings matches per-window means within 0.01 and recovers the paper's finding that embeddings shuffled within a company do as well as the real ones, and its training-free baselines on EC and MAEC reproduce all 48 published cells at the notebook's rounding. **HTML** does not: the released driver splits features and labels with three independent unseeded shuffles, and with the authors' model classes verbatim on rebuilt inputs the published recipe (ten epochs, learning rate 2×10^{-5}, raw log-volatility targets) is still converging at epoch ten; it beats the paper only at τ=3 on a random split with dropout 0, never reaches persistence at $\tau \geq 15$ on any split, and on the VolTAGE split every later paper inherited it is worse than persistence at every horizon. Its text+audio row reproduces under the shipped code's random split and under

Model	τ=3	τ=7	τ=15	τ=30
Temporal split (n=483)				
Ridge (eGeMAPS)	0.163*	0.156*	0.059*	0.079*
Ridge (WavLM)	0.156*	0.148*	-0.027	-0.008*
Ridge (emotion2vec+)	0.155*	0.129*	0.022*	0.027*
Ridge (WavLM+past-vol)	0.357	0.262	-0.044	0.052*
Ticker-disjoint split (n=484)				
Ridge (eGeMAPS)	0.203*	0.119*	0.137*	0.044*
Ridge (WavLM)	0.178*	0.192*	0.131*	0.026*
Ridge (emotion2vec+)	0.181*	0.185*	0.122*	0.047*
Ridge (WavLM+past-vol)	0.456	0.390	0.267	0.222
Audio-shuffle control of WavLM+past-vol (ticker-disjoint)				
Real audio	0.456	0.390	0.267	0.222
Within-ticker shuffle	0.453	0.373	0.229	0.152
Global shuffle	0.453	0.323	0.164	0.099

Table 8: Audio content (Stage 3). R^2_{OOS} vs. persistence on the Δv target, FinCall-Surprise (test). Audio-only heads match the train-mean intercept (0.165/0.158/0.060/0.088 temporal; 0.200/0.201/0.132/0.035 ticker-disjoint) to within 0.01; the past-vol-augmented head is below the past-vol-only ridge (Table 7) in every cell, and at $\tau \leq 7$ keeps its score when every test call receives a *random other company's* audio (global shuffle, bottom block), so past volatility, not audio content, carries it. $*$: Diebold–Mariano significant vs. the Stage-1 ticker-FE GBDT ($p < 0.05$; read direction from the value). Full head set in Appendix A.

	τ=3	τ=7	
Targets (FinCall, calls ok under both conventions)			
Sessions in calendar window (mean)	2.4	4.9	
Calls with $\geq$2 sessions (of 2,433)	1,398	2,433	
Corr($v^{\mathrm{trad}}_{\mathrm{post}}, v^{\mathrm{cal}}_{\mathrm{post}}$)	0.784	0.942	
Corr($\Delta v^{\mathrm{trad}}, \Delta v^{\mathrm{cal}}$)	0.751	0.887	
Persistence MSE, level-v, temporal test (trading / calendar)			
FinCall	1.270 / 2.071	0.448 / 0.612	0.211
MAEC	1.774 / 2.553	0.712 / 0.986	0.356
HAR-RV R^2_{OOS} vs. persistence, level-v, temporal test (trading / calendar)			
FinCall	0.444 / 0.419	0.372 / 0.402	0.138
MAEC	0.603 / 0.562	0.486 / 0.502	0.321

Table 9: Horizon convention delta. The calendar-day windows of Qin and Yang (2019) hold $\sim$0.7τ sessions; at τ=3 43% of the calls have at most one post-call session and are excluded. Persistence MSE (the leaderboard's metric) rises by 44–63% at τ=3 under the calendar convention, so published MSE values are not comparable across conventions; the FinCall temporal τ=30 deficit of HAR-RV against persistence (the COVID-regime cell, §4) holds under both conventions (-0.285 / -0.053). Full model $\times$ horizon views under both conventions: `data/results/result_table_1.tex`.

no other. **Sawhney et al.** cannot run as released (an unshipped call inventory, price files and lexicons, a TensorFlow 2.1 stack); ported, the finance SVR alone gives 0.712/0.370/0.218/0.154, the text and audio branches are untrained after the two and one epochs the scripts prescribe, and the published ensemble is reproduced because the released code tunes the mixing weights on the test set, where an untrained branch's near-constant output acts as an intercept. **DialogueGAT**'s corpus and labels were never released; ported onto FinCall's speaker turns under the paper's per-year protocol it beats persistence at short horizons and collapses on the 2020 split. **KeFVP**'s released script needed six import-level repairs before it would start; its MAEC rows use sentence embeddings regenerated with the authors' own generator because the released ones were never published, and seven of eight cells fall within one published standard deviation (the eighth, MAEC-16 at τ=15, is the row of their table that does not average to its own stated mean). On EC the released KePt embeddings are unavailable upstream (the linked folder is empty); with BERT-large pooler embeddings from the same generator, KeFVP without its knowledge-enhanced pre-training, the mean test MSE is 0.651/0.302/0.194/0.127 against the published 0.610/0.291/0.183/0.114, within one of our repeat standard deviations but above all four published means, so the row is not counted as reproduced.

Under the controls, the 30-session embargo removes half to two thirds of KeFVP's MAEC training calls, so those rows cannot separate leakage from data loss and are left to the ledger; DialogueGAT's port (one seed, 42–60 held-out calls per year) shows no consistent direction under either control.

A.4 Audio by bitrate on Earnings25

A.5 Post-cutoff evaluation under the call-date anchor

Table 14 repeats the post-cutoff evaluation with day 0 anchored on the call date on both sides of the cutoff, the anchor every released label set uses. HAR-RV barely moves, but the ridge heads that carry a learned past-volatility block collapse in-cutoff under that anchor, consistent with the block absorbing the reaction session that the after-hours rule leaves inside the pre-window for before-open calls. On Δv at τ=30 the frozen WavLM+vol head is the only one whose post-cutoff CI excludes zero under the measured anchor (0.18, CI 0.14–0.24, against HAR's 0.08), a single-quarter observation without a DM test that we report as exploratory.

A.6 Code availability

Condition (EC)	model	τ=3	τ=7	τ=15	τ=30
paper, published split, published labels	HTML (text, WWM-BERT)	1.175	0.372	0.153	0.133
published split, published labels	HTML head (BGE-M3)	0.769	0.436	0.285	0.223
	persistence	1.485	0.570	0.340	0.202
published split, our labels	HTML head (BGE-M3)	0.760	0.420	0.285	0.214
	persistence	1.826	0.590	0.333	0.183
embargoed temporal, our labels	HTML head (BGE-M3)	0.778	0.448	0.278	0.265
	persistence	1.948	0.629	0.353	0.192
ticker-disjoint, our labels	HTML head (BGE-M3)	**0.773**	**0.503**	**0.394**	**0.291**
	persistence	1.670	0.728	0.418	0.213

Table 10: Reproduced prior model under leakage-proof evaluation (Result Table 6R, first entry). The HTML head reproduces the paper's ordering against price baselines at τ=3 (and beats the published number) but not the published τ=15 and τ=30 values, which lie *below the persistence floor* of the shipped labels themselves; at τ=30 every condition is worse than persistence. Moving the same head and labels from the published split to a ticker-disjoint split raises test MSE by 2% / 20% / 38% / 36%; the embargoed temporal split costs 2% / 7% / $-$2% / 24%.

Paper / setting	what ran	ours τ=3	7	15	30	published τ=3	7	15	30
HTML (Yang et al., 2020), EC — authors' model classes verbatim, inputs rebuilt (WWM-BERT layer −2), driver corrected									
code: random 70/10/20, dropout 0	text	0.660	0.336	0.253	0.160	1.175	0.372	0.153	0.133
paper: chronological 7:1:2, dropout 0.5	text	0.972	0.439	0.406	0.248				
published split (VolTAGE), dropout 0.5	text	1.231	1.122	1.063	1.479				
persistence on the shipped labels	—	1.485	0.570	0.340	0.202				
code: random 70/10/20, dropout 0	text + 27 Praat	0.658	0.347	0.244	0.165	0.845	0.349	0.251	0.158
paper: chronological 7:1:2, dropout 0.5	text + 27 Praat	0.876	0.520	0.291	0.239				
published split (VolTAGE), dropout 0.5	text + 27 Praat	2.701	1.195	1.436	1.411				
Same-Company-Same-Signal (Yu et al., 2025), DEC — authors' `run.py`, per-window means over 19 rolling quarters									
TSMixer (price history only)	76 cells	0.561	0.252	0.210	0.257	0.561	0.252	0.211	0.256
TMLP on OpenAI embeddings	76 cells	0.608	0.285	0.215	0.202	0.587	0.283	0.211	0.205
TMLP, embeddings shuffled within ticker	76 cells	0.572	0.276	0.213	0.209	0.565	0.273	0.204	0.200
TMLP, embeddings shuffled across all	76 cells	0.646	0.315	0.259	0.249	0.637	0.319	0.255	0.247
Same-Company-Same-Signal, EC / MAEC — authors' notebook, training-free, augmented same-ticker earnings history									
EC, Aug_STPEV	no training	0.569	0.293	0.201	0.122	0.569	0.293	0.201	0.122
MAEC-15, Aug_STPEV	no training	0.443	0.214	0.130	0.112	0.443	0.214	0.130	0.112
MAEC-16, Aug_STPEV	no training	0.470	0.238	0.168	0.113	0.470	0.238	0.168	0.113
KeFVP (Niu et al., 2023) — authors' `final_series_infer.py` (six import-level release defects patched), 10 repeats									
MAEC-15, regenerated BERT-base embeddings	text + price	0.419	0.185	0.123	0.086	0.418	0.187	0.122	0.087
MAEC-16, regenerated BERT-base embeddings	text + price	0.442	0.288	0.363	0.188	0.445	0.279	0.303	0.177
EC, regenerated BERT-large embeddings (no KePt)	text + price	0.651	0.302	0.194	0.127	0.610	0.291	0.183	0.114
Sawhney et al. (2020b), EC — PyTorch port of the released scripts (their labels, split, branches); 5 seeds									
finance SVR (30 past volatilities)	port	0.712	0.370	0.218	0.154				
text BiLSTM, their 2 epochs	port	2.135	1.580	1.384	1.401				
aligned audio, their 1 epoch	port	6.514	5.295	5.368	5.130				
ensemble, (α, β) tuned on test (theirs)	port	0.639	0.298	0.175	0.102	0.601	0.308	0.181	0.119
ensemble, (α, β) tuned on validation	port	0.686	0.324	0.187	0.117				
persistence on their labels	—	1.491	0.563	0.345	0.205				
DialogueGAT (Sang and Bao, 2022) — PyG port on FinCall turns (role-typed speakers); published cells are on their unreleased corpus									
FinCall 2019 (their 2015 protocol)	port	0.694	0.320	0.206	0.134	0.453	0.324	0.190	—
persistence	—	1.429	0.566	0.289	0.139				
FinCall 2020 (their 2016 protocol)	port	0.621	0.350	0.392	0.270	0.455	0.288	0.181	—
persistence	—	1.409	0.335	0.184	0.101				
FinCall 2021 (their 2017–18 protocol)	port	0.794	0.344	0.182	0.082	0.409	0.289	0.204	—
persistence	—	1.301	0.465	0.225	0.118				

Table 11: Reproductions with the authors' own code and data (Result Table 6R-F, T6R.3). Every cell regenerates from one `ecvol reproduce` command; the fidelity ledger lists what is verbatim, patched, substituted or reason-coded per paper. Three of the five reproduce (TSMixer to 10^{-5}; TMLP within single-seed noise, including the paper's real ≈ ticker-shuffled finding; KeFVP on MAEC within one published standard deviation on seven of eight cells, with regenerated sentence embeddings; on EC, whose released KePt embeddings are unavailable upstream, the regenerated BERT-large row sits within one of our repeat standard deviations of the published means but above all four, and is not counted). HTML's text-only recipe does not reach its published long-horizon cells on any split, while its text+audio row reproduces at $\tau \geq 7$ under the shipped-code condition (random split, dropout 0) and on no other; Sawhney's published ensemble is reachable with untrained text and audio branches acting as an intercept on the finance SVR once the mixing weights are tuned on the test set, as the released code does.

Model / condition	n_{tr}	n_{te}	τ=3	7	15	30
HTML, authors' classes (EC; seeds 0–2)						
text — shipped-code random split (anchor)	≈390	112	0.660	0.336	0.253	0.160
text — embargoed	285	101	0.726	0.347	0.246	0.178
text — ticker-disjoint	386	109	0.777	0.522	0.393	0.291
persistence on the ticker-disjoint test set	—	109	1.452	0.550	0.327	0.188
text+audio — anchor	≈390	112	0.658	0.347	0.244	0.165
text+audio — embargoed	285	101	0.684	0.345	0.232	0.186
text+audio — ticker-disjoint	386	109	0.789	0.500	0.394	0.281
transcript shuffle, anchor split, the 97 test calls whose company has another call						
text — real transcript	≈390	97	0.683	0.348	0.262	0.165
text — another call of the *same* company	≈390	97	0.693	0.361	0.269	0.169
text — a call of a *different* company	≈390	97	0.745	0.386	0.318	0.203
text+audio — real / same company / different company, τ=3	≈390	97	0.675 / 0.692 / 0.784			
Sawhney et al., port — ensemble with (α, β) tuned on validation (EC; 5 seeds)						
their split (anchor)	328	110	0.686	0.324	0.187	0.117
embargoed	283	100	0.719	0.306	0.184	0.107
ticker-disjoint	383	109	0.775	0.416	0.263	0.161
its own finance SVR, ticker-disjoint	383	109	0.700	0.365	0.256	0.158
Same-Company-Same-Signal, authors' `run.py` (DEC; seed 2021; means over 19 rolling quarters)						
TMLP (text) — paired anchor	900	30	0.591	0.288	0.224	0.211
TMLP (text) — ticker-disjoint	600	30	0.660	0.354	0.280	0.267
TMLP (text) — embargoed (vs anchor 0.608 / 0.285 / 0.215 / 0.202)	895	90	0.585	0.287	0.216	0.208
TSMixer (price) — paired anchor	900	30	0.519	0.241	0.206	0.252
TSMixer (price) — ticker-disjoint	600	30	0.556	0.254	0.194	0.233
persistence on the held-out calls	—	30	1.258	0.457	0.252	0.224
KeFVP, authors' script (MAEC, regenerated BERT-base embeddings; 3 repeats)						
MAEC-15 — paired anchor	535	48	0.470	0.192	0.145	0.114
MAEC-15 — ticker-disjoint	353	48	0.458	0.199	0.150	0.113
MAEC-16 — paired anchor	980	92	0.416	0.248	0.347	0.186
MAEC-16 — ticker-disjoint	661	92	0.378	0.241	0.312	0.178
persistence on the MAEC-16 held-out calls	—	92	0.622	0.265	0.170	0.116

Table 12: The reproduced models under our controls (Result Table 6R-C, T6R.4). Each block keeps the configuration of Table 11 and changes only the split; paired anchors score the unchanged model on exactly the ticker-disjoint test calls. Removing same-company history costs the three text models 12–82% (HTML, Sawhney, TMLP; TMLP is worse in 16–18 of 19 quarters) and leaves both price-dominated models unchanged (TSMixer, KeFVP), although the ticker-disjoint training sets are as large (EC) or cut by the same third (DEC, MAEC) in both groups. The 30-session embargo moves HTML, Sawhney and SCSS by at most 13% in either direction; KeFVP's embargoed MAEC re-splits keep only a half or a third of its training calls, cannot separate leakage from data loss, and are reported in the ledger, not here. DialogueGAT (one seed, 42–60 held-out calls per year) shows no consistent direction and is also left to the ledger.

Stratum	calls (test)		τ=3	τ=7	τ=15	τ=30
All	514 (95)	real	0.435	0.328	0.171	0.220
		global shuffle	0.454	0.327	0.141	0.206
		increment	**−0.019**	**0.001**	**0.030**	**0.014**
64 kbps	281 (39)	real	0.370	0.250	0.109	0.310
		global shuffle	0.383	0.266	0.133	0.303
		increment	**−0.013**	**−0.016**	**−0.024**	**0.007**
$\leq$24 kbps	232 (55)	real	0.343	0.328	0.174	0.145
		global shuffle	0.443	0.354	0.174	0.128
		increment	**−0.100**	**−0.026**	**−0.001**	**0.018**

Table 13: Audio on Earnings25 by bitrate stratum (T9.4; R^2_{OOS} vs. persistence, Δv, ridge on WavLM + past-vol, ticker-disjoint test). The within-ticker shuffle is the identity here (one call per company), so the global shuffle is the control. Call counts are inventory calls with audio at that bitrate (one call at 32 kbps sits in neither stratum); the heads see the 475 admitted, priced calls, and the test count is their ticker-disjoint segment. The audio increment never exceeds 0.03, is negative in 7 of the 12 cells, and has no consistent sign across strata; the 64 kbps stratum is not better than the $\leq$24 kbps one. Full grid (eGeMAPS, emotion2vec+, fusion, MLP heads, all targets): `result_table_3_earnings25.csv`.

Head	seg.	τ=3	τ=7	τ=15	τ=30
HAR-RV	in	$0.44 \to 0.46$	$0.37 \to 0.41$	$0.14 \to 0.21$	$-0.29 \to -0.28$
	post	$0.60 \to 0.56$	$0.54 \to 0.51$	$0.35 \to 0.33$	$0.08 \to -0.12$
text + past-vol	in	$0.33 \to -1.71$	$0.28 \to -3.12$	$0.08 \to -3.62$	$-0.08 \to -0.47$
	post	$0.31 \to 0.28$	$0.33 \to 0.27$	$0.23 \to 0.15$	$0.12 \to -0.09$
WavLM + past-vol	in	$0.36 \to 0.26$	$0.26 \to -1.02$	$-0.04 \to -2.01$	$0.05 \to -0.56$
	post	$0.50 \to 0.43$	$0.43 \to 0.32$	$0.15 \to 0.06$	$0.18 \to 0.05$
fusion + past-vol	in	$0.29 \to -1.73$	$0.22 \to -3.13$	$0.04 \to -0.34$	$-0.11 \to -0.45$
	post	$-0.83 \to -9.97$	$0.32 \to -2.29$	$0.19 \to 0.02$	$0.01 \to -0.28$

Table 14: Anchor sensitivity: the same frozen heads with day 0 on the call date (the literature's anchor) (Δv, R^2_{OOS} vs. persistence; each cell reads measured $\to$ call-date). Ordering on level-v in the CSV.

Paper	venue	probe (date)	verdict	reason
Qin & Yang, What You Say and How You Say It Matters	ACL 2019	200, 0.0 code files, pushed 2022-08-04 (2026-09-09)	partial	dataset released (Driv
Yang et al., HTML	WWW 2020	200, 15.0 code files, pushed 2024-01-05 (2026-09-09)	partial	model classes run ve set_random_seed and dent unseeded shuffle dead Drive links — re
Sawhney et al., multimodal financial forecasting	ACM MM 2020	200, 112.0 code files, pushed 2020-10-26 (2026-09-09)	partial	TF 2.1 / Keras 2.3.1 ventory, price files an record; no multi-task Torch (T6R.3)
Sawhney et al., VolTAGE	EMNLP 2020	200, 5.0 code files, pushed 2021-11-16 (2026-09-09)	partial	ships the EC split/lab embeddings as .pkl; n
Niu et al., KeFVP	EMNLP Findings 2023	200, 42.0 code files, pushed 2024-01-28 (2026-09-09)	partial	requirements.txt does import-level repairs (names, a dropped cla imports); MAEC emb generator); EC KePt p
DialogueGAT	Findings of EMNLP 2022	200, 6.0 code files, pushed 2023-06-12 (2026-09-09)	partial	MIT code, tau<=15 or labels never released onto FinCall turns (T
Yu, Liu & He, Same Company, Same Signal	Findings of ACL 2025	200, 38.0 code files, pushed 2026-08-13 (2026-09-09)	runnable	MIT; ships DEC with mals, TSMixer 76 cel on the authors' embed
Yang et al., FinTrust	ACL 2023 short	200, 4.0 code files, pushed 2024-01-22 (2026-09-09)	partial	perturbation harness volatility model
Sound of Risk	2025	200, 0.0 code files, pushed 2025-08-26 (2026-09-09)	none	repository holds figu corpus
DeFVP	2024	409 (2026-09-09)	none	repository exists but i
ECHO-GL	2024	200, 3.0 code files, pushed 2024-02-26 (2026-09-09)	none	README states the volatility
ECC Analyzer	2024	no_url	none	no code released; GP
RiskLabs	2024	no_url	none	no code released; GP
NumHTML	AAAI 2022	no_url	none	paper promises code;
GNA-Vol		no_url	none	artefact URL returns
AMA-LSTM		no_url	none	10-byte stub repositor
AT-FinGPT		no_url	none	paywalled; no artefac

Table 15: Code-availability audit of the ECC volatility literature (T6R.2/T6R.3; regenerated by `ecvol audit code`, live GitHub probe on the date shown). One of seventeen papers ships code that runs as released; seven ship code with a broken environment, missing data, or defects that must be repaired before it starts (HTML's driver imports TensorFlow-1 symbols and decorrelates features from labels; KeFVP needs six import-level repairs); nine ship nothing usable. The reason column records what the faithful-reproduction sprint (Table 11) did with each.

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.361*	0.141	-0.367*	-1.048*
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.353*	0.244*	-0.097	-0.881*

Table 16: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.535*	0.353*	0.072	0.097
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.501*	0.440*	0.292*	0.242*

Table 17: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

A.7 Generated grids

The tables below are generated byte-identically by `ecvol report` from the committed result CSVs and are the source of truth for every curated table above. They span both corpora, the level-v and Δv targets, R^2_{OOS} and MSE, and the temporal and ticker-disjoint splits at all horizons; Result Table 1 carries both horizon conventions and the train-mean intercept, Tables 2–4 the trading-day convention with Diebold–Mariano p-values against persistence, the preceding stage and the train-mean intercept.

Stage-0/1 baselines (Result Table 1)

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.165*	0.158*	0.060	0.088*
EWMA	0.465*	0.389*	0.259*	0.019
HAR-RV	0.444*	0.372*	0.138*	-0.285*
GARCH(1,1)	0.472*	0.463*	0.298*	0.023
GBDT (ticker FE)	0.340*	0.249*	-0.126	-0.948*

Table 18: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.200*	0.201*	0.132*	0.035*
EWMA	0.520*	0.379*	0.142*	0.112*
HAR-RV	0.570*	0.491*	0.349*	0.345*
GARCH(1,1)	0.561*	0.484*	0.334*	0.298*
GBDT (ticker FE)	0.494*	0.440*	0.292*	0.237*

Table 19: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.557*	0.342*	0.024	-0.264*
EWMA	0.562*	0.389*	0.223*	0.087*
HAR-RV	0.603*	0.486*	0.321*	0.221*
GARCH(1,1)	0.596*	0.505*	0.325*	0.049
GBDT (ticker FE)	0.563*	0.429*	0.240*	0.113*

Table 20: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.511*	0.267*	0.004	-0.277*
EWMA	0.557*	0.388*	0.276*	0.073*
HAR-RV	0.574*	0.447*	0.332*	0.183*
GARCH(1,1)	0.593*	0.481*	0.403*	0.181*
GBDT (ticker FE)	0.483*	0.339*	0.221*	0.091

Table 21: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.254*	0.251*	0.187*	0.090*
EWMA	0.562*	0.389*	0.223*	0.087*
HAR-RV	0.603*	0.486*	0.321*	0.221*
GARCH(1,1)	0.596*	0.505*	0.325*	0.049
GBDT (ticker FE)	0.562*	0.420*	0.254*	0.150*

Table 22: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.265*	0.254*	0.210*	0.105*
EWMA	0.557*	0.388*	0.276*	0.073*
HAR-RV	0.574*	0.447*	0.332*	0.183*
GARCH(1,1)	0.593*	0.481*	0.403*	0.181*
GBDT (ticker FE)	0.477*	0.361*	0.250*	0.095*

Table 23: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
Train mean	0.811*	0.384	0.288*	0.235*
EWMA	0.679*	0.274*	0.156*	0.113
HAR-RV	0.706*	0.281*	0.182*	0.148*
GARCH(1,1)	0.671*	0.240*	0.148*	0.112
GBDT (ticker FE)	0.821*	0.338*	0.231	0.216*

Table 24: fincall — temporal split — level-v — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
Train mean	0.637*	0.309*	0.230	0.236
EWMA	0.657*	0.297*	0.213*	0.232*
HAR-RV	0.589*	0.243*	0.162*	0.171*
GARCH(1,1)	0.602*	0.245*	0.164*	0.184*
GBDT (ticker FE)	0.683*	0.267*	0.176*	0.198*

Table 25: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
Train mean	1.060*	0.377*	0.198	0.105*
EWMA	0.679*	0.274*	0.156*	0.113
HAR-RV	0.706*	0.281*	0.182*	0.148*
GARCH(1,1)	0.671*	0.240*	0.148*	0.112
GBDT (ticker FE)	0.838*	0.336*	0.237	0.224*

Table 26: fincall — temporal split — Δv — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
Train mean	1.095*	0.381*	0.216*	0.253*
EWMA	0.657*	0.297*	0.213*	0.232*
HAR-RV	0.589*	0.243*	0.162*	0.171*
GARCH(1,1)	0.602*	0.245*	0.164*	0.184*
GBDT (ticker FE)	0.692*	0.267*	0.176*	0.200*

Table 27: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
Train mean	0.787*	0.469*	0.347	0.253*
EWMA	0.778*	0.435*	0.276*	0.183*
HAR-RV	0.705*	0.366*	0.242*	0.156*
GARCH(1,1)	0.717*	0.352*	0.240*	0.191
GBDT (ticker FE)	0.775*	0.406*	0.270*	0.178*

Table 28: maec — temporal split — level-v — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
Train mean	0.836*	0.463*	0.324	0.206*
EWMA	0.756*	0.387*	0.236*	0.149*
HAR-RV	0.730*	0.350*	0.218*	0.132*
GARCH(1,1)	0.695*	0.328*	0.194*	0.132*
GBDT (ticker FE)	0.884*	0.417*	0.254*	0.147

Table 29: maec — ticker_disjoint split — level-v — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
Train mean	1.323*	0.533*	0.289*	0.183*
EWMA	0.778*	0.435*	0.276*	0.183*
HAR-RV	0.705*	0.366*	0.242*	0.156*
GARCH(1,1)	0.717*	0.352*	0.240*	0.191
GBDT (ticker FE)	0.778*	0.413*	0.265*	0.171*

Table 30: maec — temporal split — Δv — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
Train mean	1.256*	0.471*	0.257*	0.144*
EWMA	0.756*	0.387*	0.236*	0.149*
HAR-RV	0.730*	0.350*	0.218*	0.132*
GARCH(1,1)	0.695*	0.328*	0.194*	0.132*
GBDT (ticker FE)	0.894*	0.404*	0.244*	0.146*

Table 31: maec — ticker_disjoint split — Δv — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.376*	0.234*	-0.079	-0.718*
EWMA	0.404*	0.420*	0.328*	0.145*
HAR-RV	0.419*	0.402*	0.275*	-0.053
GARCH(1,1)	0.388*	0.484*	0.399*	0.145*
GBDT (ticker FE)	0.249*	0.278*	0.105	-0.368*

Table 32: fincall — temporal split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.503*	0.486*	0.206*	-0.015
EWMA	0.470*	0.476*	0.258*	0.036
HAR-RV	0.534*	0.559*	0.418*	0.289*
GARCH(1,1)	0.520*	0.546*	0.409*	0.262*
GBDT (ticker FE)	0.472*	0.498*	0.340*	0.213*

Table 33: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.062	0.205*	0.104*	0.064*
EWMA	0.404*	0.420*	0.328*	0.145*
HAR-RV	0.419*	0.402*	0.275*	-0.053
GARCH(1,1)	0.388*	0.484*	0.399*	0.145*
GBDT (ticker FE)	0.225*	0.270*	0.101	-0.389*

Table 34: fincall — temporal split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.124*	0.235*	0.189*	0.075*
EWMA	0.470*	0.476*	0.258*	0.036
HAR-RV	0.534*	0.559*	0.418*	0.289*
GARCH(1,1)	0.520*	0.546*	0.409*	0.262*
GBDT (ticker FE)	0.464*	0.503*	0.319*	0.212*

Table 35: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.508*	0.412*	0.198*	-0.148
EWMA	0.566*	0.412*	0.311*	0.169*
HAR-RV	0.562*	0.502*	0.400*	0.265*
GARCH(1,1)	0.579*	0.497*	0.428*	0.203*
GBDT (ticker FE)	0.513*	0.450*	0.357*	0.172*

Table 36: maec — temporal split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.626*	0.379*	0.139*	-0.153
EWMA	0.653*	0.445*	0.337*	0.172*
HAR-RV	0.661*	0.498*	0.396*	0.233*
GARCH(1,1)	0.663*	0.515*	0.452*	0.284*
GBDT (ticker FE)	0.575*	0.416*	0.292*	0.139*

Table 37: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.247*	0.245*	0.224*	0.135*
EWMA	0.566*	0.412*	0.311*	0.169*
HAR-RV	0.562*	0.502*	0.400*	0.265*
GARCH(1,1)	0.579*	0.497*	0.428*	0.203*
GBDT (ticker FE)	0.515*	0.442*	0.370*	0.186*

Table 38: maec — temporal split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.297*	0.281*	0.229*	0.154*
EWMA	0.653*	0.445*	0.337*	0.172*
HAR-RV	0.661*	0.498*	0.396*	0.233*
GARCH(1,1)	0.663*	0.515*	0.452*	0.284*
GBDT (ticker FE)	0.576*	0.420*	0.319*	0.144*

Table 39: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.071	0.612	0.298	0.152
Train mean	1.293*	0.469*	0.322	0.262*
EWMA	1.234*	0.355*	0.200*	0.130*
HAR-RV	1.204*	0.366*	0.216*	0.161
GARCH(1,1)	1.271*	0.316*	0.179*	0.130*
GBDT (ticker FE)	1.555*	0.442*	0.267	0.209*

Table 40: fincall — temporal split — level-v — MSE (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.550	0.738	0.321	0.220
Train mean	1.268*	0.379*	0.255*	0.223
EWMA	1.352*	0.387*	0.238*	0.212
HAR-RV	1.189*	0.325*	0.186*	0.156*
GARCH(1,1)	1.227*	0.334*	0.188*	0.162*
GBDT (ticker FE)	1.347*	0.371*	0.212*	0.173*

Table 41: fincall — ticker_disjoint split — level-v — MSE (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.071	0.612	0.298	0.152
Train mean	1.942	0.486*	0.267*	0.143*
EWMA	1.234*	0.355*	0.200*	0.130*
HAR-RV	1.204*	0.366*	0.216*	0.161
GARCH(1,1)	1.271*	0.316*	0.179*	0.130*
GBDT (ticker FE)	1.606*	0.447*	0.268	0.212*

Table 42: fincall — temporal split — Δv — MSE (test) — calendar-day τ (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.550	0.738	0.321	0.220
Train mean	2.233*	0.565*	0.260*	0.203*
EWMA	1.352*	0.387*	0.238*	0.212
HAR-RV	1.189*	0.325*	0.186*	0.156*
GARCH(1,1)	1.227*	0.334*	0.188*	0.162*
GBDT (ticker FE)	1.366*	0.367*	0.218*	0.173*

Table 43: fincall — ticker_disjoint split — Δv — MSE (test) — calendar-day τ (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.553	0.986	0.492	0.263
Train mean	1.256*	0.579*	0.395*	0.302
EWMA	1.109*	0.579*	0.339*	0.219*
HAR-RV	1.119*	0.491*	0.295*	0.194*
GARCH(1,1)	1.075*	0.496*	0.282*	0.210*
GBDT (ticker FE)	1.243*	0.542*	0.317*	0.218*

Table 44: maec — temporal split — level-v — MSE (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.494	0.910	0.426	0.229
Train mean	0.933*	0.565*	0.366*	0.264
EWMA	0.867*	0.505*	0.282*	0.189*
HAR-RV	0.845*	0.457*	0.258*	0.175*
GARCH(1,1)	0.840*	0.441*	0.233*	0.164*
GBDT (ticker FE)	1.059*	0.531*	0.302*	0.197*

Table 45: maec — ticker_disjoint split — level-v — MSE (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.553	0.986	0.492	0.263
Train mean	1.924*	0.744*	0.382*	0.228*
EWMA	1.109*	0.579*	0.339*	0.219*
HAR-RV	1.119*	0.491*	0.295*	0.194*
GARCH(1,1)	1.075*	0.496*	0.282*	0.210*
GBDT (ticker FE)	1.237*	0.550*	0.310*	0.214*

Table 46: maec — temporal split — Δv — MSE (test) — calendar-day τ (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Persistence	2.494	0.910	0.426	0.229
Train mean	1.753*	0.654*	0.328*	0.193*
EWMA	0.867*	0.505*	0.282*	0.189*
HAR-RV	0.845*	0.457*	0.258*	0.175*
GARCH(1,1)	0.840*	0.441*	0.233*	0.164*
GBDT (ticker FE)	1.056*	0.528*	0.290*	0.196*

Table 47: maec — ticker_disjoint split — Δv — MSE (test) — calendar-day τ (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.430*	0.322*	-0.007	-0.539*
MLP (text)	0.153*	-0.235*	-1.156*	-2.782*
Ridge (past-vol)	0.407*	0.314*	0.079*	-0.382*
MLP (past-vol)	0.439*	0.352*	0.116*	-0.459*
Ridge (text+vol)	0.435*	0.339*	0.034*	-0.454*
MLP (text+vol)	0.219*	-0.115*	-0.918*	-2.504*

Table 48: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.549*	0.449	0.254	0.199
MLP (text)	0.177*	-0.484*	-1.491*	-1.354*
Ridge (past-vol)	0.543	0.467	0.341	0.284
MLP (past-vol)	0.571*	0.512*	0.365*	0.327
Ridge (text+vol)	0.552*	0.464	0.293	0.278
MLP (text+vol)	0.209*	-0.348*	-1.277*	-1.166*

Table 49: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

Stage-2 text heads (Result Table 2)

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.182*	0.202	0.044*	-0.133*
MLP (text)	0.013*	0.069*	-0.253	-1.202
Ridge (past-vol)	0.412*	0.314*	0.094*	0.113*
MLP (past-vol)	0.441*	0.330*	0.086*	-0.401*
Ridge (text+vol)	0.333	0.285	0.080*	-0.077*
MLP (text+vol)	0.275	0.222	-0.108	-0.855

Table 50: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.195*	0.217*	0.195*	0.169
MLP (text)	0.046*	-0.024*	-0.108*	-0.061*
Ridge (past-vol)	0.552*	0.469	0.345	0.318*
MLP (past-vol)	0.577*	0.508*	0.381*	0.356*
Ridge (text+vol)	0.399*	0.375	0.256	0.301
MLP (text+vol)	0.423*	0.352*	0.165*	0.224

Table 51: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.515	0.304*	0.002*	-0.303*
MLP (text)	-0.174*	-1.142*	-2.602*	-5.867*
Ridge (past-vol)	0.575	0.441	0.322*	0.242*
MLP (past-vol)	0.605*	0.470	0.324*	0.240*
Ridge (text+vol)	0.516	0.348*	0.165	0.010*
MLP (text+vol)	-0.116*	-1.036*	-1.853*	-4.494*

Table 52: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.556*	0.355	0.116	-0.115*
MLP (text)	0.233*	-0.523*	-1.583*	-3.321*
Ridge (past-vol)	0.530	0.364	0.230	0.159
MLP (past-vol)	0.537	0.389	0.239	0.184
Ridge (text+vol)	0.563*	0.390	0.274	0.133
MLP (text+vol)	0.103*	-0.670*	-1.556*	-3.281*

Table 53: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.247*	0.148*	0.191	0.098
MLP (text)	0.110*	0.003*	-0.287*	-0.288*
Ridge (past-vol)	0.581	0.444	0.323*	0.242*
MLP (past-vol)	0.615*	0.494*	0.349*	0.258*
Ridge (text+vol)	0.438*	-0.015*	0.233	0.152
MLP (text+vol)	0.407*	0.291*	0.109*	-0.174*

Table 54: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.276*	0.267*	0.221	0.113
MLP (text)	0.245*	0.177*	0.070*	-0.112*
Ridge (past-vol)	0.530	0.364	0.230	0.159
MLP (past-vol)	0.559*	0.436*	0.318*	0.182
Ridge (text+vol)	0.453	0.340	0.235	0.093
MLP (text+vol)	0.457	0.322	0.168	-0.070*

Table 55: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.724*	0.304*	0.212	0.177*
MLP (text)	1.075*	0.553*	0.455*	0.434*
Ridge (past-vol)	0.752*	0.307*	0.194*	0.159*
MLP (past-vol)	0.713*	0.290*	0.186*	0.168*
Ridge (text+vol)	0.718*	0.296*	0.204*	0.167*
MLP (text+vol)	0.992*	0.499*	0.404*	0.402*

Table 56: fincall — temporal split — level-v — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.617*	0.263	0.185	0.210
MLP (text)	1.125*	0.708*	0.619*	0.616*
Ridge (past-vol)	0.626	0.254	0.163	0.187
MLP (past-vol)	0.587*	0.233*	0.158*	0.176
Ridge (text+vol)	0.613*	0.256	0.176	0.189
MLP (text+vol)	1.082*	0.643*	0.565*	0.567*

Table 57: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.039*	0.357	0.202*	0.130*
MLP (text)	1.253*	0.417*	0.264	0.253
Ridge (past-vol)	0.746*	0.307*	0.191*	0.102*
MLP (past-vol)	0.710*	0.300*	0.193*	0.161*
Ridge (text+vol)	0.847	0.320	0.194*	0.124*
MLP (text+vol)	0.921	0.348	0.234	0.213

Table 58: fincall — temporal split — Δv — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.101*	0.374*	0.200*	0.217
MLP (text)	1.306*	0.489*	0.275*	0.278*
Ridge (past-vol)	0.613*	0.254	0.163	0.178*
MLP (past-vol)	0.578*	0.235*	0.154*	0.168*
Ridge (text+vol)	0.822*	0.298	0.185	0.183
MLP (text+vol)	0.790*	0.309*	0.207*	0.203

Table 59: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.861	0.495*	0.355*	0.261*
MLP (text)	2.082*	1.525*	1.281*	1.377*
Ridge (past-vol)	0.755	0.398	0.241*	0.152*
MLP (past-vol)	0.701*	0.377	0.240*	0.152*
Ridge (text+vol)	0.859	0.464*	0.297	0.198*
MLP (text+vol)	1.981*	1.450*	1.015*	1.102*

Table 60: maec — temporal split — level-v — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	0.758*	0.408	0.288	0.180*
MLP (text)	1.310*	0.962*	0.841*	0.697*
Ridge (past-vol)	0.802	0.402	0.251	0.136
MLP (past-vol)	0.791	0.386	0.248	0.132
Ridge (text+vol)	0.746*	0.385	0.236	0.140
MLP (text+vol)	1.532*	1.055*	0.833*	0.690*

Table 61: maec — ticker_disjoint split — level-v — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.336*	0.607*	0.288	0.181
MLP (text)	1.578*	0.710*	0.458*	0.258*
Ridge (past-vol)	0.744	0.396	0.241*	0.152*
MLP (past-vol)	0.684*	0.360*	0.231*	0.149*
Ridge (text+vol)	0.997*	0.723*	0.273	0.170
MLP (text+vol)	1.052*	0.505*	0.317*	0.235*

Table 62: maec — temporal split — Δv — MSE (test) (n=529).

Model	τ=3	τ=7	τ=15	τ=30
Ridge (text)	1.237*	0.463*	0.254	0.143
MLP (text)	1.291*	0.520*	0.303*	0.179*
Ridge (past-vol)	0.802	0.402	0.251	0.136
MLP (past-vol)	0.753*	0.357*	0.222*	0.132
Ridge (text+vol)	0.934	0.417	0.249	0.146
MLP (text+vol)	0.929	0.428	0.271	0.172*

Table 63: maec — ticker_disjoint split — Δv — MSE (test) (n=532).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.361	0.144*	-0.370*	-1.159*
mlp eGeMAPS	-10.085	-20.164	-38.738	-61.570
ridge eGeMAPS +vol	0.374	0.260	-0.067	-0.677
mlp eGeMAPS +vol	-7.831	-17.515	-37.949	-50.038
ridge WavLM	0.357	0.084*	-0.507*	-1.106*
mlp WavLM	-0.072*	-0.808*	-2.505*	-5.253*
ridge WavLM +vol	0.378	0.245	-0.094	-0.665*
mlp WavLM +vol	0.016*	-0.582*	-1.882*	-4.053*
ridge emotion2vec+	0.354	0.152*	-0.320*	-0.967
mlp emotion2vec+	-0.270*	-1.582*	-3.692*	-6.470*
ridge emotion2vec+ +vol	0.396	0.305*	0.078*	0.398*
mlp emotion2vec+ +vol	-0.313*	-1.580*	-3.559*	-6.632*
ridge WavLM+eGeMAPS	0.354	0.139*	-0.411*	-1.114*
mlp WavLM+eGeMAPS	-1.847	-5.053	-9.675	-14.344
ridge WavLM+eGeMAPS +vol	0.367	0.259	-0.119	-0.763
mlp WavLM+eGeMAPS +vol	-2.005	-5.365	-10.910	-16.691
ridge WavLM+text +vol	0.427*	0.332*	-0.007	-0.527*
mlp WavLM+text +vol	0.213*	-0.050*	-0.823*	-2.165*

Table 64: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=
ridge eGeMAPS	0.535	0.320*	0.073*	0.06
mlp eGeMAPS	-7.712	-26.589	-52.220	-53.9
ridge eGeMAPS +vol	0.544	0.426	0.310	0.2
mlp eGeMAPS +vol	-7.149	-21.128	-47.690	-46.8
ridge WavLM	0.514	0.320*	-0.022*	-0.0
mlp WavLM	-0.020*	-0.939*	-2.346*	-2.2
ridge WavLM +vol	0.521	0.401	0.243	0.2
mlp WavLM +vol	-0.074*	-1.072*	-2.573*	-2.3
ridge emotion2vec+	0.533	0.357*	0.085*	0.1
mlp emotion2vec+	-1.783	-4.890	-10.950	-9.6
ridge emotion2vec+ +vol	0.540	0.463	0.335	0.2
mlp emotion2vec+ +vol	-1.860	-5.254	-11.588	-11.1
ridge WavLM+eGeMAPS	0.516	0.305*	-0.071*	-0.0
mlp WavLM+eGeMAPS	-1.442	-4.849	-10.335	-9.3
ridge WavLM+eGeMAPS +vol	0.522	0.389	0.230	0.2
mlp WavLM+eGeMAPS +vol	-1.676	-5.476	-11.476	-10.3
ridge WavLM+text +vol	0.539	0.448	0.276	0.2
mlp WavLM+text +vol	0.228*	-0.288*	-1.033*	-0.86

Table 65: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

Stage-3 audio heads (Result Table 3)

FinCall-Surprise only (MAEC ships no raw audio).

Model	τ=3	τ=7	τ=15	τ=
ridge eGeMAPS	0.163*	0.156*	0.059*	0.07
mlp eGeMAPS	-0.096*	-0.004*	-0.228	-0.5
ridge eGeMAPS +vol	0.373	0.268	0.074*	0.10
mlp eGeMAPS +vol	0.319	-0.212	-0.901	-3.6
ridge WavLM	0.156*	0.148*	-0.027	-0.00
mlp WavLM	0.045*	-0.108*	-0.433*	-1.1
ridge WavLM +vol	0.357	0.262	-0.044	0.05
mlp WavLM +vol	0.295	0.087*	-0.387*	-1.27
ridge emotion2vec+	0.155*	0.129*	0.022*	0.02
mlp emotion2vec+	0.006*	-0.029*	-0.337*	-1.0
ridge emotion2vec+ +vol	0.403*	0.307	0.095*	0.06
mlp emotion2vec+ +vol	0.186*	0.107*	-0.218	-0.9
ridge WavLM+eGeMAPS	0.155*	0.145*	0.015*	-0.00
mlp WavLM+eGeMAPS	-0.010*	-0.180*	-0.550*	-1.1
ridge WavLM+eGeMAPS +vol	0.353	0.268	0.033*	0.05
mlp WavLM+eGeMAPS +vol	0.140	-0.014*	-0.581*	-1.49
ridge WavLM+text +vol	0.311	0.237	0.050*	-0.10
mlp WavLM+text +vol	0.323	0.269	-0.103	-0.9

Table 66: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.203*	0.119*	0.137*	0.044*
mlp eGeMAPS	0.093*	-0.037*	0.038*	-0.004*
ridge eGeMAPS +vol	0.524	0.431	0.201	0.138*
mlp eGeMAPS +vol	0.328	0.169*	0.179*	0.139*
ridge WavLM	0.178*	0.192*	0.115*	0.025*
mlp WavLM	0.026*	0.011*	-0.140*	0.119*
ridge WavLM +vol	0.456	0.390	0.267	0.232
mlp WavLM +vol	0.424*	0.346*	0.139*	0.050*
ridge emotion2vec+	0.181*	0.185*	0.122*	0.047*
mlp emotion2vec+	0.077*	0.032*	-0.188*	0.210*
ridge emotion2vec+ +vol	0.541*	0.464	0.340	0.317
mlp emotion2vec+ +vol	0.431*	0.323*	0.123*	0.029*
ridge WavLM+eGeMAPS	0.181*	0.194*	0.136*	0.041*
mlp WavLM+eGeMAPS	0.033*	-0.007*	-0.131*	0.119*
ridge WavLM+eGeMAPS +vol	0.442	0.364*	0.244	0.217
mlp WavLM+eGeMAPS +vol	0.327*	0.154*	-0.152*	0.136*
ridge WavLM+text +vol	0.369*	0.348*	0.228	0.201
mlp WavLM+text +vol	0.404*	0.315*	0.128*	0.103*

Table 67: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.636	0.325*	0.230*	0.246*
mlp eGeMAPS	11.919	13.169	13.212	14.372
ridge eGeMAPS +vol	0.624	0.274	0.171	0.192
mlp eGeMAPS +vol	11.149	10.562	12.088	12.512
ridge WavLM	0.664	0.325*	0.254*	0.266*
mlp WavLM	1.396*	0.926*	0.831*	0.861*
ridge WavLM +vol	0.655	0.286	0.188	0.197
mlp WavLM +vol	1.470*	0.989*	0.887*	0.882*
ridge emotion2vec+	0.639	0.307*	0.227*	0.231*
mlp emotion2vec+	3.808	2.812	2.967	2.799
ridge emotion2vec+ +vol	0.630	0.256	0.165	0.188
mlp emotion2vec+ +vol	3.912	2.985	3.125	3.171
ridge WavLM+eGeMAPS	0.663	0.332*	0.266*	0.270*
mlp WavLM+eGeMAPS	3.340	2.792	2.814	2.700
ridge WavLM+eGeMAPS +vol	0.653	0.291	0.191	0.199
mlp WavLM+eGeMAPS +vol	3.661	3.091	3.097	2.975
ridge WavLM+text +vol	0.631	0.263	0.180	0.191
mlp WavLM+text +vol	1.056*	0.615*	0.505*	0.487*

Table 69: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	0.811	0.383*	0.289*	0.248*
mlp eGeMAPS	14.075	9.472	8.379*	7.196*
ridge eGeMAPS +vol	0.795	0.331	0.225	0.193*
mlp eGeMAPS +vol	11.213	8.287	8.213*	5.862*
ridge WavLM	0.816	0.410*	0.318*	0.242*
mlp WavLM	1.361*	0.809*	0.739*	0.718*
ridge WavLM +vol	0.789	0.338	0.231	0.191*
mlp WavLM +vol	1.250*	0.708*	0.608*	0.580*
ridge emotion2vec+	0.821	0.380*	0.278*	0.226
mlp emotion2vec+	1.612*	1.156*	0.989*	0.858*
ridge emotion2vec+ +vol	0.766	0.311*	0.196*	0.161*
mlp emotion2vec+ +vol	1.667*	1.154*	0.961*	0.877*
ridge WavLM+eGeMAPS	0.820	0.386*	0.298*	0.243*
mlp WavLM+eGeMAPS	3.615	2.709	2.251*	1.763*
ridge WavLM+eGeMAPS +vol	0.804	0.332	0.236	0.203*
mlp WavLM+eGeMAPS +vol	3.815	2.849	2.511*	2.033*
ridge WavLM+text +vol	0.727*	0.299*	0.212*	0.175*
mlp WavLM+text +vol	1.000*	0.470*	0.384*	0.364*

Table 68: fincall — temporal split — level-v — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	1.062*	0.378*	0.198*	0.106*
mlp eGeMAPS	1.391*	0.449*	0.259	0.180
ridge eGeMAPS +vol	0.797	0.328	0.195*	0.102*
mlp eGeMAPS +vol	0.865	0.543	0.401	0.535
ridge WavLM	1.072*	0.381*	0.217	0.116*
mlp WavLM	1.212*	0.496*	0.302*	0.248
ridge WavLM +vol	0.817	0.330	0.220	0.109*
mlp WavLM +vol	0.895	0.409*	0.292*	0.261*
ridge emotion2vec+	1.074*	0.390*	0.206*	0.112*
mlp emotion2vec+	1.263*	0.461*	0.282*	0.230
ridge emotion2vec+ +vol	0.757*	0.310	0.191*	0.108*
mlp emotion2vec+ +vol	1.033*	0.400*	0.257	0.229
ridge WavLM+eGeMAPS	1.073*	0.383*	0.208*	0.116*
mlp WavLM+eGeMAPS	1.283*	0.528*	0.327*	0.244
ridge WavLM+eGeMAPS +vol	0.822	0.327	0.204*	0.109*
mlp WavLM+eGeMAPS +vol	1.091	0.454*	0.333*	0.286*
ridge WavLM+text +vol	0.875	0.342	0.200*	0.127*
mlp WavLM+text +vol	0.860	0.327	0.233	0.228

Table 70: fincall — temporal split — Δv — MSE (test) (n=483).

Model	τ=3	τ=7	τ=15	τ=30
ridge eGeMAPS	1.091*	0.420*	0.214*	0.250*
mlp eGeMAPS	1.241*	0.495*	0.239*	0.273*
ridge eGeMAPS +vol	0.652	0.272	0.176	0.187
mlp eGeMAPS +vol	0.920	0.397*	0.204*	0.212
ridge WavLM	1.125*	0.386*	0.216*	0.255*
mlp WavLM	1.333*	0.472*	0.283*	0.313*
ridge WavLM +vol	0.745	0.291	0.182	0.204
mlp WavLM +vol	0.788*	0.312*	0.214*	0.246*
ridge emotion2vec+	1.121*	0.389*	0.218*	0.249*
mlp emotion2vec+	1.263*	0.462*	0.295*	0.317*
ridge emotion2vec+ +vol	0.628*	0.256	0.164	0.179
mlp emotion2vec+ +vol	0.778*	0.323*	0.218*	0.254*
ridge WavLM+eGeMAPS	1.120*	0.385*	0.214*	0.233*
mlp WavLM+eGeMAPS	1.323*	0.481*	0.281*	0.312*
ridge WavLM+eGeMAPS +vol	0.764	0.304*	0.188	0.205
mlp WavLM+eGeMAPS +vol	0.921*	0.404*	0.286*	0.313*
ridge WavLM+text +vol	0.863*	0.311*	0.189	0.183
mlp WavLM+text +vol	0.815*	0.327*	0.217*	0.211

Table 71: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.361	0.141	-0.367	-1.048
HAR-RV	0.444	0.372	0.138	-0.285
GBDT (ticker FE)	0.353	0.244	-0.097	-0.881
Text (BGE+vol)	0.435	0.339	0.034	-0.454
Audio (WavLM+eG+vol)	0.367	0.259	-0.119	-0.763
Fusion (stack)	0.442	0.390	0.164	-0.227

Table 72: fincall — temporal split — level-v — R^2_OOS vs persistence (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.535	0.353	0.072	0.097
HAR-RV	0.570	0.491	0.349	0.345
GBDT (ticker FE)	0.501	0.440	0.292	0.242
Text (BGE+vol)	0.552	0.464	0.293	0.278
Audio (WavLM+eG+vol)	0.522	0.389	0.230	0.239
Fusion (stack)	0.556	0.466	0.319	0.365

Table 73: fincall — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=484).

Consolidated ablation grid (Result Table 4)

One pre-registered canonical model per stage; $*$ marks Holm-corrected significance vs. Stage-1 within the Δv confirmatory family (two-sided; read direction from the value).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.165	0.158	0.060	0.088
HAR-RV	0.444	0.372	0.138	-0.285
GBDT (ticker FE)	0.340	0.249	-0.126	-0.948
Text (BGE+vol)	0.333	0.285	0.080*	-0.077*
Audio (WavLM+eG+vol)	0.353	0.268	0.033*	0.050*
Fusion (stack)	0.406*	0.347*	0.066*	0.100*

Table 74: fincall — temporal split — Δv — R^2_OOS vs persistence (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.200	0.201	0.132	0.035
HAR-RV	0.570	0.491	0.349	0.345
GBDT (ticker FE)	0.494	0.440	0.292	0.237
Text (BGE+vol)	0.399*	0.375	0.256	0.301
Audio (WavLM+eG+vol)	0.442	0.364	0.241	0.217
Fusion (stack)	0.531	0.465	0.340	0.396*

Table 75: fincall — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.557	0.342	0.024	-0.264
HAR-RV	0.603	0.486	0.321	0.221
GBDT (ticker FE)	0.563	0.429	0.240	0.113
Text (BGE+vol)	0.516	0.348	0.165	0.010
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 76: maec — temporal split — level-v — R^2_OOS vs persistence (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.511	0.267	0.004	-0.277
HAR-RV	0.574	0.447	0.332	0.183
GBDT (ticker FE)	0.483	0.339	0.221	0.091
Text (BGE+vol)	0.563	0.390	0.274	0.133
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 77: maec — ticker_disjoint split — level-v — R^2_OOS vs persistence (test) (n=532).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.254	0.251	0.187	0.090
HAR-RV	0.603	0.486	0.321	0.121
GBDT (ticker FE)	0.562	0.420	0.254	0.130
Text (BGE+vol)	0.438*	-0.015*	0.233	0.152
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 78: maec — temporal split — Δv — R^2_OOS vs persistence (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
Train mean	1.060	0.377	0.198	0.105
HAR-RV	0.706	0.281	0.182	0.148
GBDT (ticker FE)	0.838	0.336	0.237	0.224
Text (BGE+vol)	0.847	0.320	0.194*	0.124*
Audio (WavLM+eG+vol)	0.822	0.327	0.204*	0.109*
Fusion (stack)	0.755*	0.292*	0.197*	0.103*

Table 82: fincall — temporal split — Δv — MSE (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	0.000	0.000	0.000	0.000
Train mean	0.265	0.254	0.210	0.105
HAR-RV	0.574	0.447	0.332	0.181
GBDT (ticker FE)	0.477	0.361	0.250	0.095
Text (BGE+vol)	0.453	0.340	0.235	0.093
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 79: maec — ticker_disjoint split — Δv — R^2_OOS vs persistence (test) (n=532).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
Train mean	1.095	0.381	0.216	0.253
HAR-RV	0.589	0.243	0.162	0.171
GBDT (ticker FE)	0.692	0.267	0.176	0.200
Text (BGE+vol)	0.822*	0.298	0.185	0.183
Audio (WavLM+eG+vol)	0.764	0.304	0.188	0.205
Fusion (stack)	0.642	0.255	0.164	0.158*

Table 83: fincall — ticker_disjoint split — Δv — MSE (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.270	0.448	0.211	0.115
Train mean	0.811	0.384	0.288	0.235
HAR-RV	0.706	0.281	0.182	0.148
GBDT (ticker FE)	0.821	0.338	0.231	0.216
Text (BGE+vol)	0.718	0.296	0.204	0.167
Audio (WavLM+eG+vol)	0.804	0.332	0.236	0.203
Fusion (stack)	0.708	0.273	0.176	0.141

Table 80: fincall — temporal split — level-v — MSE (test) (n=483).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
Train mean	0.787	0.469	0.347	0.253
HAR-RV	0.705	0.366	0.242	0.156
GBDT (ticker FE)	0.775	0.406	0.270	0.178
Text (BGE+vol)	0.859	0.464	0.297	0.198
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 84: maec — temporal split — level-v — MSE (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.368	0.477	0.248	0.262
Train mean	0.637	0.309	0.230	0.236
HAR-RV	0.589	0.243	0.162	0.171
GBDT (ticker FE)	0.683	0.267	0.176	0.198
Text (BGE+vol)	0.613	0.256	0.176	0.189
Audio (WavLM+eG+vol)	0.653	0.291	0.191	0.199
Fusion (stack)	0.607	0.255	0.169	0.166

Table 81: fincall — ticker_disjoint split — level-v — MSE (test) (n=484).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
Train mean	0.836	0.463	0.324	0.206
HAR-RV	0.730	0.350	0.218	0.132
GBDT (ticker FE)	0.884	0.417	0.254	0.147
Text (BGE+vol)	0.746	0.385	0.236	0.140
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 85: maec — ticker_disjoint split — level-v — MSE (test) (n=532).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.774	0.712	0.356	0.201
Train mean	1.323	0.533	0.289	0.183
HAR-RV	0.705	0.366	0.242	0.156
GBDT (ticker FE)	0.778	0.413	0.265	0.171
Text (BGE+vol)	0.997*	0.723*	0.273	0.170
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 86: maec — temporal split — Δv — MSE (test) (n=529).

Stage	τ=3	τ=7	τ=15	τ=30
Persistence	1.709	0.632	0.326	0.161
Train mean	1.256	0.471	0.257	0.144
HAR-RV	0.730	0.350	0.218	0.132
GBDT (ticker FE)	0.894	0.404	0.244	0.146
Text (BGE+vol)	0.934	0.417	0.249	0.146
Audio (WavLM+eG+vol)	—	—	—	—
Fusion (stack)	—	—	—	—

Table 87: maec — ticker_disjoint split — Δv — MSE (test) (n=532).

Substrate audit and reproduction grids

The per-file substrate audit behind Table 1 is released as `result_table_5r.csv` and `result_table_5r_labels.csv` with the lineage mirror manifest; the reproduction grids as `result_table_6r*.csv` (HTML on EC with three seeds and persistence per cell; Same-Company-Same-Signal on DEC per rolling quarter; the faithful runs and their controls per paper).